

Solved: Software Engineering Assignments

Comprehensive Solutions for Assignments 1-10

Assignment 1: SDLC Models Comparison

Model	Best For	Example
Waterfall	Fixed requirements	Govt. Infrastructure projects
Agile	Changing requirements	Mobile App development
Spiral	High-risk projects	Military/Aerospace software

Assignment 2: SRS snippet (Library System)

Functional Req: System must allow users to search books by ISBN. System must track return deadlines.

Non-Functional Req: System should support 100 concurrent users with < 2s response time.

Assignment 3: UML Diagrams (Student System)

Use Case: Actors: Student, Admin. Actions: Register Course, View Grades, Manage Records.

Class Diagram: Classes: Student (id, name), Course (code, title), Enrollment (date, grade).

Assignment 4: Testing & Test Cases

Unit: Testing individual functions. **Integration:** Testing combined modules.

Sample Test Case: Login Page - Input: Correct Creds; Expected: Redirect to Dashboard; Result: Pass.

Assignment 5: Agile & Scrum Plan

Sprint Duration: 2 Weeks. **Sprint Backlog:** User Authentication, Database Schema setup, UI Prototype.

Assignment 8: Version Control (Git)

- `git init`: Initialize local repo
- `git add .`: Stage changes
- `git commit -m "msg"`: Save snapshot
- `git push`: Upload to GitHub

Assignment 10: Architecture Patterns

MVC: Model (Data), View (UI), Controller (Logic). Example: Django or Laravel apps.

Microservices: Breaking app into small independent services. Example: Netflix, Amazon.